Steroidogenesis

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Block: Endocrine/Urologic/Reproductive

Thread: Biochemistry

Conflict of Interest: Dr. Root has no conflicts of interest to disclose.

Learning Objectives

By exam time, you should be able to do the following:

1. Map the pathways involved in synthesis of adrenal steroids and explain how tissue-specific production of corticosteroids is achieved.
2. Describe the hormonal regulation of adrenal steroid production and explain why chronic signaling leads to congenital adrenal hyperplasia (CAH).
3. Map the pathways involved in the synthesis of testosterone, estrogen and progesterone in gonadal tissue and describe their hormonal regulation.
4. Compare and contrast the effects on blood pressure, metabolism, stress response, and sexual development of common disorders of steroidogenesis (21-hydroxylase deficiency, 11β-hydroxylase deficiency, 17α-hydroxylase deficiency, 11β-hydroxysteroid dehydrogenase deficiency) and describe their pathophysiological mechanisms.
5. Outline common laboratory tests for identifying patients with defects in steroidogenesis and explain how they discriminate between 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

Additional Resources

Biason-Lauber A. (2003) Disorders of Steroid Synthesis and Metabolism. In: Blau N., Duran M., Blaskovics M.E., Gibson K.M. (eds). *Physician's Guide to the Laboratory Diagnosis of Metabolic Diseases*. Springer, Berlin, Heidelberg

Bremer AA, Miller WL (2014) Regulation of Steroidogenesis. In: Ulloa-Aguirre A, Conn PM (eds). *Cellular Endocrinology in Health and Disease*. Academic Press.

Miller WL, Auchus RJ (2011) The Molecular Biology, Biochemistry, and Physiology of Human Steroidogenesis and Its Disorders. *Endocrine Reviews* 32, 81-151.

Study Questions

1. How is tissue-specific production of aldosterone, cortisol and DHEA achieved in the adrenal cortex?

2. Compare and contrast the hormonal regulation of aldosterone, cortisol, and DHEA production in the adrenal cortex.

3. By what biochemical mechanisms do angiotensin II and ACTH upregulate aldosterone and cortisol production in the short term (minutes)? Describe what happens with long-term stimulation of these hormones.

4. Describe the synthesis and regulation of testosterone, estrogen and progesterone in the testes and ovaries.

5. Compare and contrast the pathogenesis related to mineralocorticoid, glucocorticoid and androgen effects in 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

6. Summarize how 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency can be distinguished through laboratory testing.

Case 1:

A 10-day-old infant presented to the Emergency Department with severe electrolyte imbalance and life-threatening arrhythmia. The parents reported a 3-day history of poor feeding and lethargy. On examination, he was bradycardic (heart rate of 65 beats/min) with signs of dehydration. His ECG showed broad complex bradycardia. Blood gas showed metabolic acidosis with hyponatraemia and hyperkalaemia.

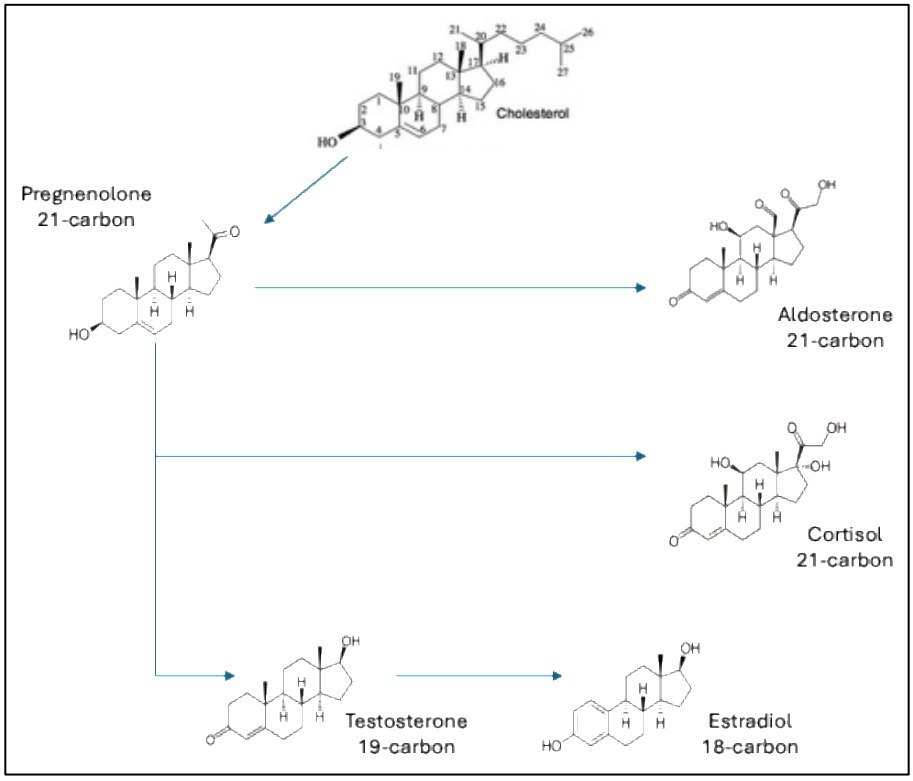
(BMJ Case Rep. 2019; 12(1): e227565; doi: 10.1136/bcr-2018-227565)

Question: What common congenital abnormality might explain severe electrolyte imbalance, dehydration and metabolic acidosis in a newborn?

Answer: A defect in adrenal steroid biosynthesis

A. Steroids

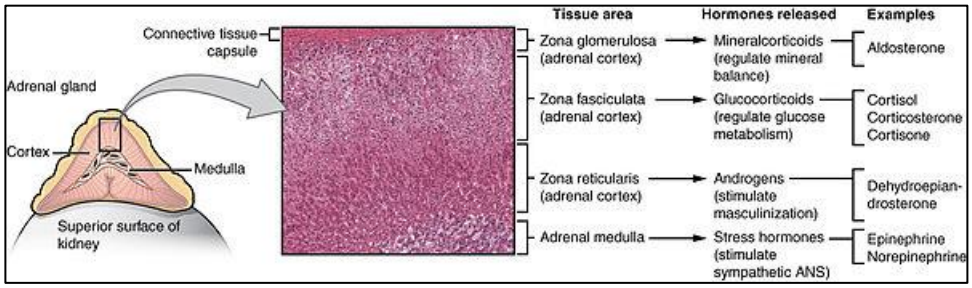
1. Steroids are cell-permeable hormones derived from cholesterol that bind to and activate type I nuclear hormone receptors (*see NHR lecture*).
2. Four types:
   - **Mineralocorticoids** – important for electrolyte and blood pressure regulation (e.g., aldosterone)
   - **Glucocorticoids** – important for the stress response, glucose handling, metabolism regulation and the inflammatory response (e.g., cortisol)
   - **Androgens** – important for development and maintenance of the penis, testes and prostate gland, for growth of muscle and bone, for stimulation of facial, axillary and pubic hair, and for other pleiotropic anabolic effects (e.g., testosterone)
   - **Estrogens** – important for growth and development of the uterus, vagina and breast, for regulation of menarche, for distribution of adipose tissue, for stimulation of axillary and pubic hair, and for the growth of bones (e.g., estradiol)



3. Expression patterns of enzymes and cofactors/enzyme modifiers determine tissue specific steroid production
4. Steroid intermediates of these reactions can diffuse between cells to complete synthesis

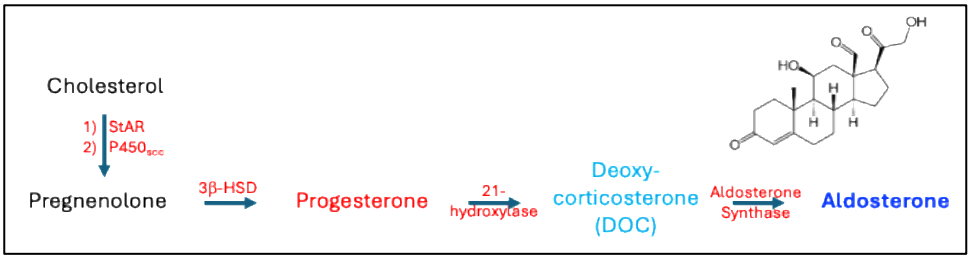
5. By definition, steroidogenic tissues can synthesize steroids de novo from cholesterol (distinguishing steroidogenic tissue from other tissues that merely modify steroid hormones).
6. The first steps of steroidogenesis, common to all pathways, involve transporting cholesterol to mitochondria where the acyl chain is removed:
    a. **Steroidogenic acute regulatory (StAR) protein** transports membrane anchored cholesterol through the cytoplasm to the mitochondrial inner membrane. The activity of StAR is generally the rate-limiting step of steroid synthesis and can be modified by signal transduction cascades (through phosphorylation).
    b. **Cholesterol side chain cleavage (SCC) enzyme** residing in the inner mitochondrial membrane cleaves the acyl chain, converting membrane anchored cholesterol (27-carbons) to the membrane-permeable steroid intermediate pregnenolone (21-carbons). Due to their membrane permeability, steroid hormones (intermediates and end products) cannot be stored in steroidogenic tissue, and regulation of steroid levels must be achieved by the rate of *de novo* synthesis.

B. Adrenal Steroid Synthesis



1. Mineralocorticoid synthesis

Mineralocorticoids bind to mineralocorticoid receptors (MR) in renal tubule epithelia to activate expression of the sodium/potassium pump in the baso-lateral membrane and sodium and potassium channels in the luminal membrane, stimulating resorption of sodium (and, by proxy, water) to increase blood volume at the expense of increased potassium excretion



a. Step 1 –

Cholesterol side-chain cleavage (SCC)

b. Step 2 – **3-beta-hydroxysteroid dehydrogenase (3βHSD)**

- Common to **mineralocorticoid and glucocorticoid** synthesis
- **Progesterone** is an antagonist of the MR (potassium-sparing diuretics (spironolactones) are analogs of progesterone)

c. Step 3 – **21-hydroxylase**

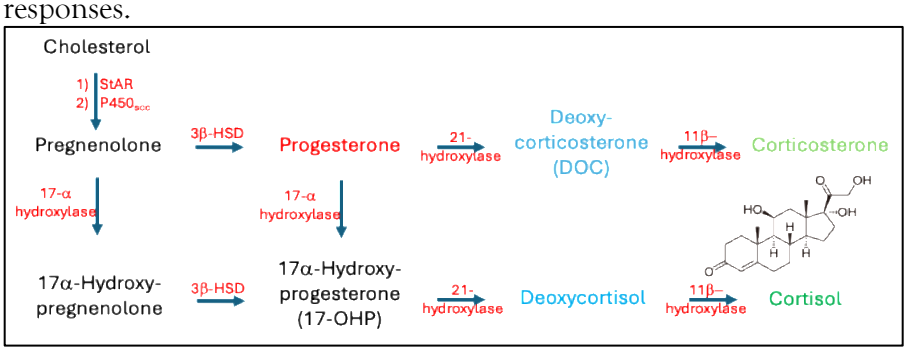
- Common to **mineralocorticoid and glucocorticoid** synthesis
- **Deoxycorticosterone (DOC)** has mineralocorticoid activity

d. Step 4 – **Aldosterone synthase**

- Attaches hydroxyl (-OH) to carbon 11 and oxygen (=O) to carbon 18
- **Expressed exclusively in the zona glomerulosa;** generates the **unique** chemical feature of aldosterone **(C18 aldehyde)**
- Most of the binding affinity of aldosterone for MR comes from the hydroxyl at carbon 11 (which will also be found in cortisol)
- ***NOTE:*** *the enzymes that hydroxylate position 11 in aldosterone (aldosterone synthase) and cortisol (11-βhydroxylase) are* **different gene products expressed in different zones of the adrenal cortex!** *The presentation of these pathways in First Aid misleads some students.*

2. **Glucocorticoid synthesis**

Glucocorticoids bind to glucocorticoid receptors (GR) in a wide variety of tissues to promote gluconeogenesis & lipolysis and inhibit inflammatory responses.



a. The zona fasciculata expresses two steroid biosynthesis enzymes that are not expressed in the zona glomerulosa: **17α-hydroxylase** and **11β-hydroxylase**.

b. 17α-hydroxylation is characteristic of glucocorticoids and a significant determinant of GR binding affinity and specificity.

c. The zona fasciculata can synthesize most intermediates produced in the

zona glomerulosa, **but cannot make aldosterone** because only aldosterone synthase, which is expressed exclusively in the zona glomerulosa, can generate the C18 aldehyde.

d. Since steroids are membrane permeant, intermediates synthesized in the zona glomerulosa (e.g., pregnenolone, progesterone) can follow the direction of blood flow (outside to inside) to supplement intermediates in the zona fasciculata. This explains, in part, why **cortisol is 1000 times more concentrated in serum than aldosterone**, and why adrenal androgens are the most abundant corticosteroids.

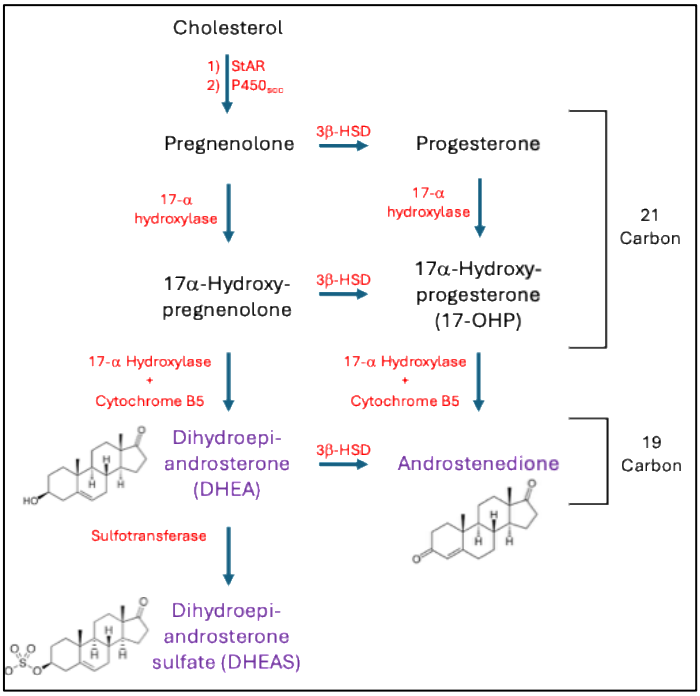
e. In addition to its primary role as a glucocorticoid, cortisol binds tightly to and activates MR. **Specificity for aldosterone is enforced by expression of 11β-hydroxysteroid dehydrogenase (11βHSD) in the distal nephron (and placenta). 11βHSD catalyzes oxidation of cortisol (11-OH) to cortisone (11=O), which does not bind to the mineralocorticoid receptor.** The C18 aldehyde makes aldosterone a poor 11βHSD substrate. Mutations to 11βHSD cause a severe form of hypertension (apparent mineralocorticoid excess (AME)) that is unresponsive to RAAS (rare).

3. Adrenal androgen synthesis

Adrenal androgens bind to the androgen receptor (AR) to induce anabolic changes in many systems, support development of the penis, testes and prostate gland, and stimulate facial, axillary and pubic hair growth.

a. Step 1 - Cholesterol side-chain cleavage (SCC)
b. Step 2 – 17α -hydroxylase
   - Attaches hydroxyl to carbon 17
   - Common in zona fasciculata (cortisol) and zona reticularis

c. Step 3 - 17α -hydroxylase **+ cytochrome b5**
   - Oxidation of carbon 17 to ketone that also removes carbons 20 and 21, converting the 21-carbon steroid 17α-pregnenolone to a 19-carbon androgen **dehydroxyepiandrosterone (DHEA)**
   - Cytochrome b5 is a cofactor for 17α-hydroxylase and is exclusively expressed in the zona reticularis (and other androgen and estrogen-producing steroidogenic tissues)



d. Step 4 – **Sulfotransferase**

- Forms **dehydroxyepiandrosterone sulfate (DHEAS**, see below)

e. Expression of 3βHSD in the zona reticularis enables synthesis of **androstendione**.

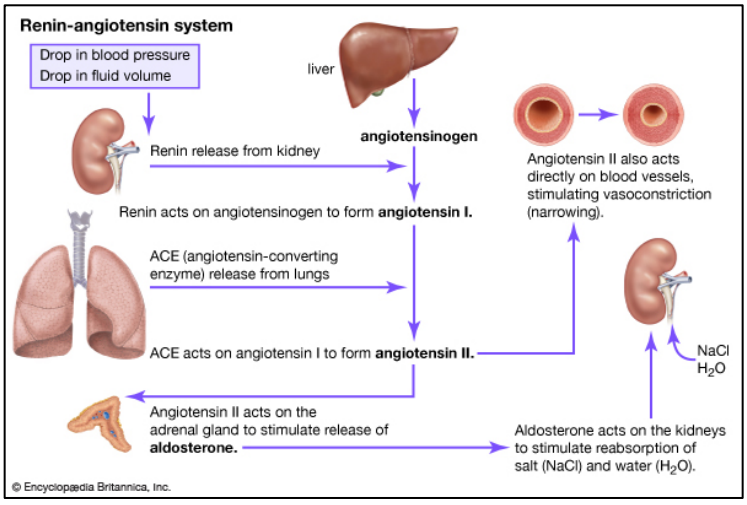
f. Properties of adrenal androgens

- DHEA and androstenedione have low affinity for AR (androstendione > DHEA), but both can be converted to strong androgens (testosterone and DHT) in peripheral tissues (e.g., skin, hair follicles, prostate).
- DHEA has weak estrogen receptor (ER) activity but can be transformed into potent estrogens in peripheral tissues (e.g. vagina).
- Due to the activity of sulfotransferase in the zona reticularis, DHEAS is produced at 100x the levels of DHEA. It has greater solubility and can bind better to albumin, granting it 10-20-fold longer serum half-life than DHEA.
- DHEAS does not bind AR but is readily converted by sulfatases to DHEA in peripheral tissues.
- DHEAS is the most abundant circulating steroid in prepubescent and post-menopausal individuals. Adrenals produce around 15 mg per day in healthy young adults.
- DHEA/DHEAS levels vary throughout life:

- High in neonate: fetus and newborns do not have a zona reticularis. Fetal adrenal glands produce androgens from the **fetal zone** under stimulation of ACTH from fetal pituitary gland. The fetal zone involutes at birth, leaving behind cortex with zonae glomerulosa and fasciculata. Rapid loss of fetal zone accounts for precipitous fall of androgen levels at birth.
- Low to absent in early childhood
- Rises at age 7-8 due to an unknown stimulus. The increase in adrenal androgen production heralds in the period of adrenarche and leads to the development of secondary sexual characteristics such as pubic and axillary hair, oily skin and body odor. Adrenarche precedes the onset of full puberty.
- Peaks at 20 to 30 years of age**,** with a gradual decline thereafter.
- DHEA/S is the source of most estrogens in post-menopausal women.

C. Regulation of Adrenal Steroid Synthesis

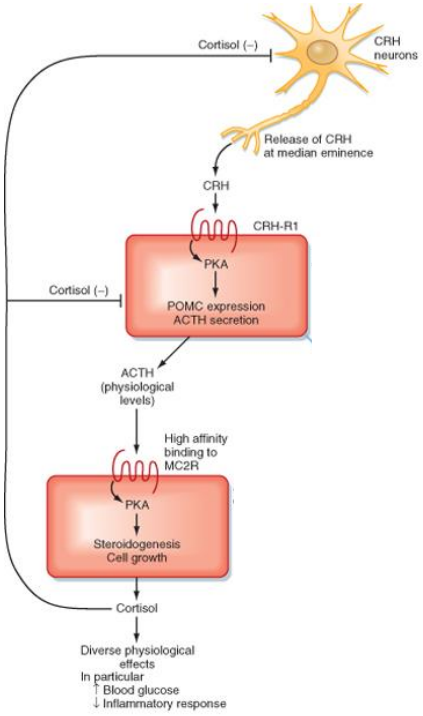
1. **Mineralocorticoid** production in the adrenal zona glomerulosa is regulated by the renin-angiotensin-aldosterone system (RAAS).
   - Drop in BP increases renal secretion of renin



   - Renin converts angiotensinogen to angiotensin I
   - Pulmonary ACE converts angiotensin I to angiotensin II
   - Angiotensin II binds to angiotensin II receptors (AGTR1, Gq-coupled GPCR) to stimulate aldosterone production in the zona glomerulosa

- Feedback is through changes in blood pressure mediated by increased sodium retention and vasoconstriction

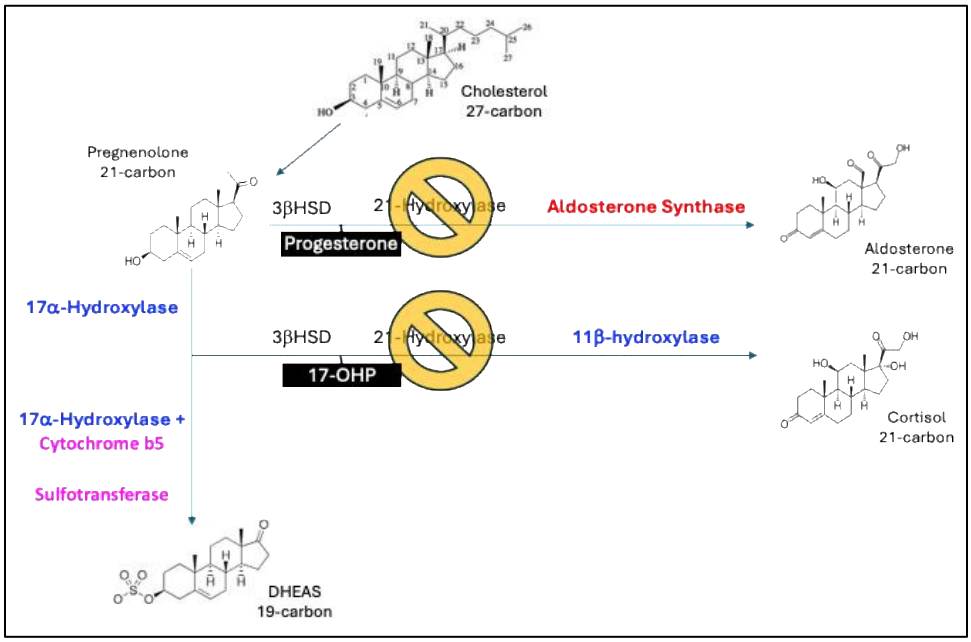
2. **Glucocorticoid** production in the adrenal zona fasciculata is regulated by the hypothalamus-pituitary-adrenal (HPA) axis.
   - The hypothalamus secretes corticotropin releasing hormone (CRH) in response to a variety of neural and hormonal inputs
   - CRH binds to the CRH receptor in the anterior pituitary to stimulate secretion of adrenocorticotropic hormone (ACTH)
   - ACTH binds receptors (ACTHR, Gs-coupled GPCR) in the zona fasciculata that regulate cortisol production
   - Cortisol negatively feeds back on the hypothalamus and pituitary to reduce CRH and ACTH secretion



3. The regulation of **adrenal androgen** production in the zona reticularis remains a mystery. However, this tissue does express the ACTHR and is responsive to ACTH signaling. However, unlike cortisol, DHEA does **NOT** exert feedback control over CRH or ACTH secretion.

4. Receptor stimulation of adrenal cortical tissues leads to:
   - Short-term (minutes): PKC (through Gq activation) or PKA (through Gs activation) phosphorylate StAR, increasing the rate of cholesterol transport to the mitochondrial inner membrane.
   - Medium-term (hours to days): increases cellular levels of enzymes and cofactors involved in steroid-hormone synthesis.
   - Long-term (days to weeks): increases cell mass of activated zone through proliferative pathways (e.g., MAPK).
     - Unregulated, long-term stimulation of Gs and/or Gq-coupled GPCRs in the adrenal cortex activates proliferative pathways (e.g., the mitogen-activated protein kinase (MAPK) pathway), causing **congenital adrenal hyperplasia (CAH)**.

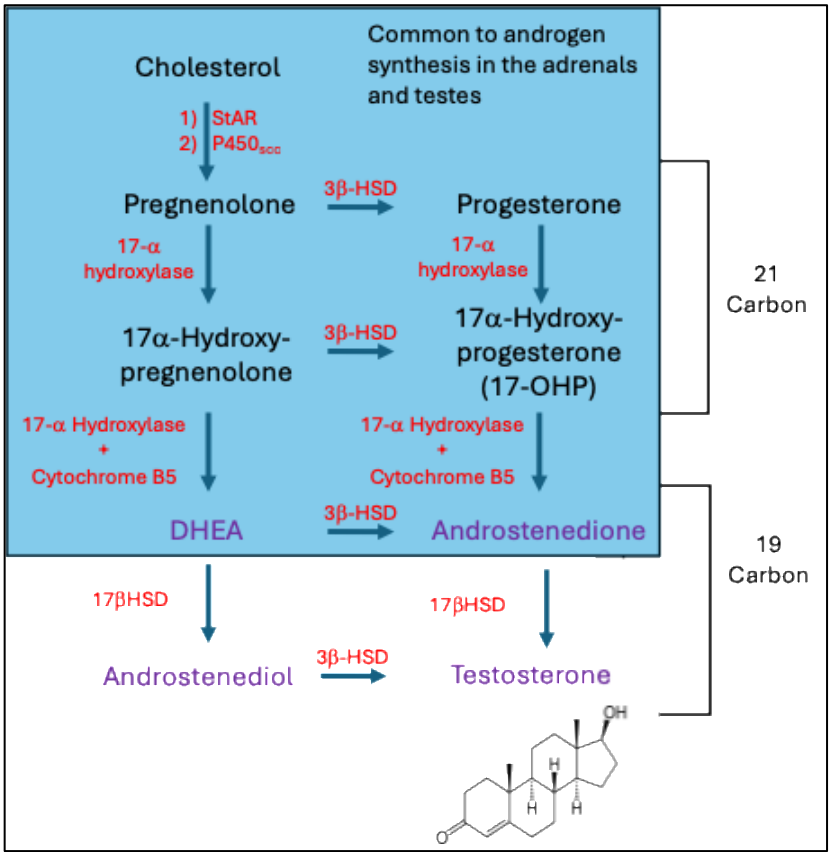
D. Clinical Interlude: 21-hydroxylase deficiency



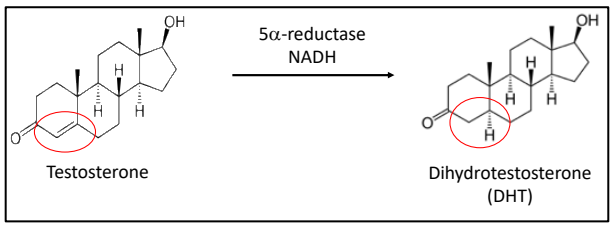
1. Impaired production of aldosterone
   - Electrolyte imbalance: hyponatremia, hyperkalemia
   - Hypovolemic hypotension
   - Chronically elevated Angiotensin II
2. Impaired production of cortisol
   - Reduced stress response and impaired metabolism
   - Lethargy
   - Chronically elevated ACTH
3. Increased secretion of progesterone (MR antagonist) and 17-0HP, intermediate steroids on the synthesis pathway to aldosterone and cortisol
4. Increased levels of androstenedione due to progesterone and 17-OHP release from zonae glomerulus and fasciculata feeding into the steroidogenic pathway in the zona reticularis
   - Impaired secondary sexual development (e.g., virilization of XX female)
5. Increased cell mass of all zonae due to long term stimulation of AGTR1 (zona glomerulosa) and ACTHR (zonae fasciculata and reticularis) → congenital adrenal hyperplasia.
   - The high ACTH levels and hyperplasia makes the amount of adrenal androgens produced even more pronounced

E. Gonadal androgens and estrogens

1. **Testosterone** synthesis in the testes
   - Follows the same synthetic pathway as that for adrenal androgens
   - One additional step: **17β-hydroxysteroid dehydrogenase (17βHSD)** – reduces the ketone on C17 to a hydroxyl



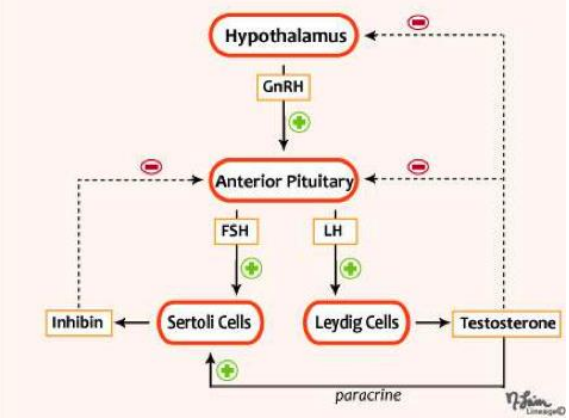
2. Testosterone conversion to **Dihydrotestosterone (DHT)**
   - DHT has higher affinity for AR
   - Conversion carried out by **5α-Reductase**



   - 2 isoforms – one in peripheral tissue (e.g., hair follicles) and another predominantly in reproductive tissue (e.g., prostate)

- Also utilized in the degradative pathway for cortisol, progesterone, 17-OHP and other 21-carbon steroids
- Inhibitors of 5α-reductase used for treatment of benign prostate hyperplasia and androgenic alopecia (patterned hair loss)
- 5α–Reductase deficiency
  - Incidence: rare (<1 in 100,000 births), but can account for up to 10% of cases of DSD in some ethnicities (internal urogenital tract, pseudovagina)

3. Hormonal regulation of androgen synthesis in testes

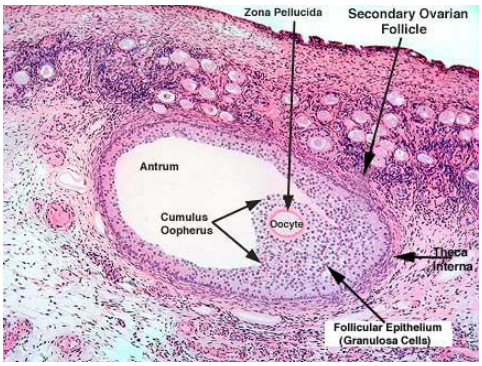


- **Luteinizing hormone (LH) and Follicle-Stimulating hormone (FSH)**
  - Hypothalamus secretes gonadotropin-releasing hormone (GnRH) in response to variety of neural and hormonal inputs.
  - GnRH binds to GnRH receptor (a GPCR coupled to Gq) in anterior pituitary to stimulate secretion of LH and FSH
  - LH binds receptors (GPCR coupled to Gs) in testicular Leydig cells regulating testosterone production
  - Testosterone feeds back on the hypothalamus and anterior pituitary gland to negatively regulate GNRH and LH secretion in XY individuals. Interestingly, this feedback regulation is primarily mediated through estrogen signaling from the conversion of testosterone to estradiol by the enzyme aromatase expressed in neurons and gonadotrophs (see E.4).
  - FSH binds to receptors (GPCR coupled to Gs) in testicular Sertoli cells (which support spermatogenesis), leading to secretion of inhibin which negatively feeds back on FSH secretion
- LH and FSH receptor activation leads to signaling through cAMP and PKA, increasing StAR activity (short term) and ultimately increasing cell proliferation (long term)

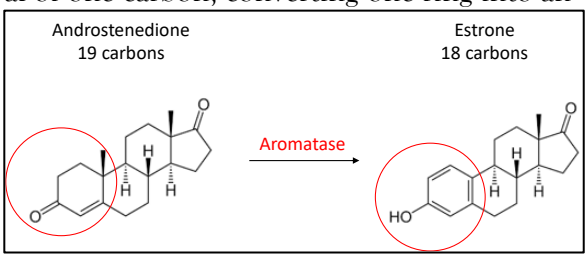
4. **Estrogen** synthesis in ovaries

   Estrogens bind to the estrogen receptor (ER) to induce growth and development of the uterus, vagina and breast, regulate menarche, and stimulate axillary and pubic hair growth.

   - Theca cells and granulosa cells in the ovary partake together in the synthesis of estrogens (primarily estradiol). During the **follicular phase** of the menstrual cycle, **theca cells** organize on the periphery of the follicle, while **granulosa cells** form the follicular barrier and surround the oocyte.



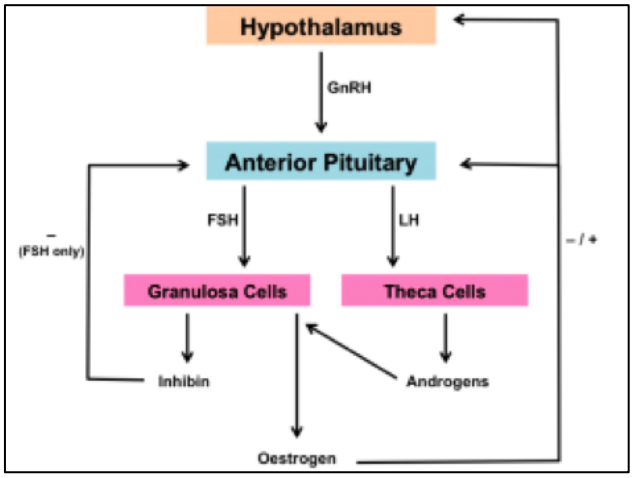
   - During follicular stage of the menstrual cycle, theca cells produce androgens (primarily **androstenedione**) through the same pathway used in the adrenal zona reticularis and testicular Leydig cells (although theca cells express low levels of 17βHSD and, therefore, do not make a lot of testosterone).
   - During follicular stage of the menstrual cycle, granulosa cells convert androstenedione (and low amounts of testosterone) into estrogens **estrone** and **estradiol**. Activated granulosa cells contain high levels of:
     - 17βHSD
     - **Aromatase**
       - Catalyzes removal of one carbon, converting one ring into an aromatic ring.



       - Expressed in steroidogenic tissues (ovaries, placenta), in brain, and in non-steroidogenic tissue (bone, fat)
         - Aromatase in placenta is critical to protect mother from exposure to fetal androgens
         - Aromatase in brain assists the neuroprotective effects of estrogen (shown to protect CNS from damage following spinal cord injuries, traumatic brain injuries, ischemic brain injuries in animal models)

- Aromatase in bone is critical for **epiphyseal maturation**
  - Epiphyseal plate is the layer of hyaline cartilage at the ends of long bones where longitudinal bone growth takes place
  - Epiphyseal maturation occurs at the end of puberty where all the cartilage is replaced by bone.
- Aromatase inhibitors are used to treat gynecomastia and hormone-sensitive breast cancer in postmenopausal patients.
- Aromatase deficiency
  - Incidence: rare (<1 in 100,000 births)
  - Androgen accumulation leads to XX virilization and DSD
  - Failure of epiphyseal maturation leads to tall stature (XX & XY)

5. Hormonal regulation of estrogen and progesterone in the ovaries
   - Estrogen synthesis is regulated by LH and FSH in much the same way testosterone synthesis is regulated in the testes



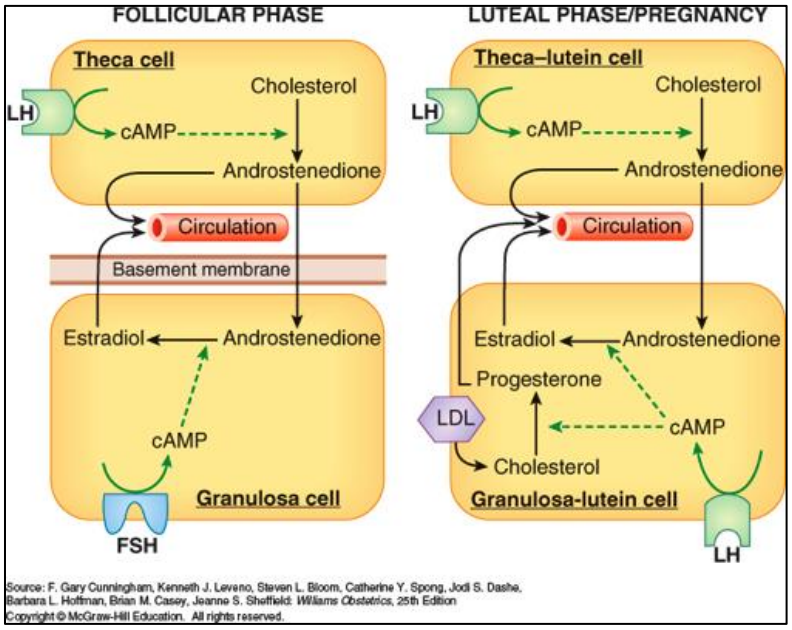
   - GnRH from the hypothalamus stimulates the anterior pituitary to release LH and FSH.
   - During follicular phase, LH stimulates the theca cells to produce androgens (primarily androstenedione) from cholesterol (much like LH stimulates androgen production in Leydig cells in the testes).
   - During follicular phase, FSH stimulates granulosa cells to convert androgens to estrogens by upregulation of aromatase. High levels of FSH also induce the expression of inhibin and the LH receptor.
   - Estrogens regulate GnRH, LH and FSH secretion from the hypothalamus and pituitary, but their actual effect (stimulatory or inhibitory) depends on the phase of the menstrual cycle (more

information on this will be presented in reproductive physiology lectures).

- Inhibin negatively feeds back on FSH secretion.

- After ovulation, the follicular cells transform into theca-lutein cells and granulosa lutein cells.
  - Theca-lutein cells stimulated by LH produce androgens from cholesterol.
  - Granulosa-lutein cells, stimulated now by both LH and FSH, produce estrogens from androgens (derived fom theca-lutein cells) and begin to produce progesterone from cholesterol due to the induced expression of StAR and 3βHSD.

F.



Source: F. Gary Cunningham, Kenneth J. Leveno, Steven L. Bloom, Catherine Y. Spong, Jodi S. Dashe, Barbara L. Hoffman, Brian M. Casey, Jeanne S. Sheffield: *Williams Obstetrics*, 25th Edition
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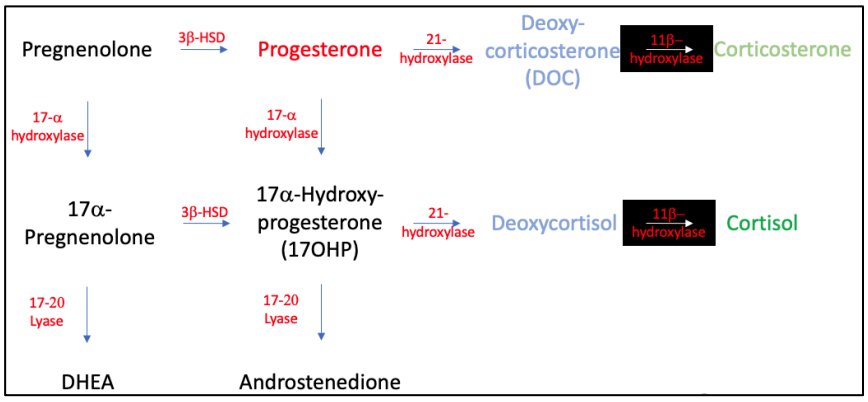
Disorders of Steroidogenesis

1. **21-hydroxylase deficiency**
    - Incidence: 1:5000 to 1:15000 live births
    - **Most common cause of congenital adrenal hyperplasia (CAH)**
    - Caused by mutations in or deletion of 21-hydroxylase gene
        - Disease requires homozygous or compound heterozygous mutations (both alleles must be affected)
    - Three forms depending on residual enzymatic activity in steroidogenic tissue
        - Salt-wasting – Enzymatic activity <1% of normal. This is an endocrinological emergency due to inability to produce aldosterone – diagnosed and treated at birth
        - Simple virilizing – Enzymatic activity <5% of normal. Recognized at birth due to difference in sexual differentiation (DSD) in XX offspring (production of adrenal androgens is stimulated by 21-hydroxylase deficiency because lack of cortisol production results in chronically elevated ACTH)
        - Cryptic or late-onset – often not diagnosed until childhood or early adulthood
    - For detailed pathogenesis, see **Clinical Correlation in Section D** above.
        - Salt-wasting form
            - Aldosterone and cortisol production are compromised, leading to increased levels of ACTH (in utero) and AGII and ACTH (after birth). Both contribute to CAH. (Note that maternal aldosterone is sufficient to protect in utero; salt wasting develops postpartum.)
            - Hyponatremia, hyperkalemia, hypotension, hypoglycemia, decreased epinephrine levels, lethargy, failure to thrive in neonate
            - Increased ACTH and increased production of progesterone and 17-OHP lead to increased production of androstenedione from fetal adrenals→virilization of XX external genitalia.
        - Simple-virilizing form
            - Cortisol production is compromised, leading to increased levels of ACTH → CAH.
            - Hypoglycemia, decreased epinephrine levels, lethargy, failure to thrive

- High androgen levels resulting in virilization of XX external genitalia.

Case 1 (*cont*): With karyotype and hormonal results, virilized female and salt depleting 21-hydroxylase deficiency were diagnosed. Treatment with hydrocortisone, mineralocorticoid and salt replacement was started. …deletion was detected in the CYP21A2 gene of the patient in line with 21-hydroxylase and salt wasting type.

2. **11β-hydroxylase deficiency**
   - Incidence: 1:100,000 live births (**7% of CAH cases**)
   - Caused by mutations in or deletion of 11β-hydroxylase gene
     - Disease requires homozygous or compound heterozygous mutations (both alleles must be affected)
   - No impact on the ability of the adrenal zona glomerulosa to produce aldosterone (*NOTE*: 11β-hydroxylation of aldosterone is catalyzed by aldosterone synthase, which is a completely different enzyme!) but **renin levels are extremely low** (due to unregulated mineralocorticoid activity of DOC and deoxycortisol), so **aldosterone levels are low.**



   - Impact on the ability of the adrenal zona fasciculata to produce cortisol **AND** corticosterone
     - **Serum cortisol and corticosterone ↓**
       - **Hypoglycemia**
     - **ACTH ↑** (causes CAH)
     - **Serum DOC and deoxycortisol ↑** (mineralocorticoid activity)
       - **Serum sodium ↑**

- **Serum potassium ↓**
- Hypertension (*low renin levels*)

- Adrenal androgen production **INCREASED** (virilizing) due to effect of elevated ACTH on zona reticularis and presence of elevated levels of steroid precursors from zona fasciculata

Case 2: A 40-year-old…patient presented with…lack of breast development, complete absence of axillary and pubic hair, tall stature, and slenderness. The external genitalia were phenotypically female, the height was 175 cm, the weight was 64 Kg, and the body mass index (BMI) was 20.9. On admission, the blood pressure (BP) was persistently high throughout the day: mean systolic BP values were 170 mmHg; mean diastolic BP values were 110 mmHg. The patient received biochemical and endocrine lab tests. The results…showed the following: low-undetectable levels of testosterone, DHEA-S, androstenedione, 17-OH-progesterone, estradiol and active renin; low-normal 24-h urinary cortisol and serum aldosterone; high ACTH, progesterone, LH and FSH concentrations. Chromosomal analysis demonstrated a…46,XY, karyotype.

Papi, *et al., Case Rep. Endo.* (2018); https://doi.org/10.1155/2018/2086861

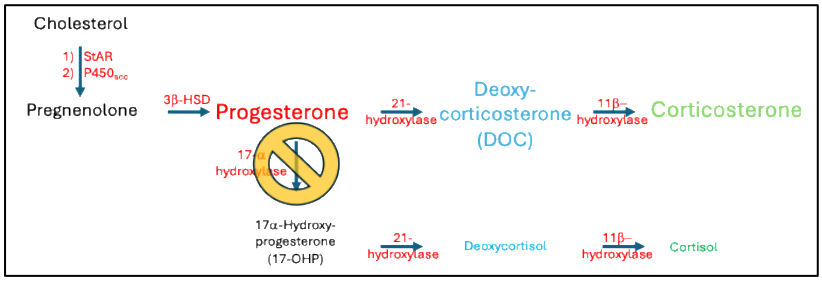
Question: What is responsible for atypical development of secondary sexual characteristics in this patient?

3. **17α-hydroxylase deficiency**
   - Incidence
     - Complete absence of enzymatic activity – rare (< 1:100,000)
     - Incomplete loss of enzymatic activity may account for a significant percentage of primary hypertension (by GWAS analysis)
   - Caused by mutations in or deletion of 17α-hydroxylase gene
     - Disease requires homozygous or compound heterozygous mutations (both alleles must be affected)
   - No impact on the ability of the adrenal zona glomerulosa to produce aldosterone, but….

     Renin levels are extremely low (due to unregulated mineralo-corticoid

activity of DOC)**, so aldosterone levels are low**

- Impact on the ability of the adrenal zona fasciculata to produce cortisol



- **Serum cortisol ↓**
- **ACTH ↑** (causes CAH)
- **Serum DOC ↑** (mineralocorticoid activity)
  - **Serum sodium ↑**
  - **Serum potassium ↓**
  - **Hypertension** (low renin levels)
- **Serum corticosterone ↑** (partially compensates for cortisol deficiency, so glucose is low to normal)
- Impaired androgen and estrogen production
  - **Low androgen/estrogen levels** due to defects in DHEA/androstenedione synthesis (need 17α-hydroxylase)
  - XX infants appear normal; pathology arises in puberty
  - **XY infants show absent or incomplete sexual development** (hypospadias, micropenis, intra-abdominal or inguinal testes)

G. Testing for CAH

1. Newborns in the US are tested for congenital adrenal hyperplasia at birth by determining the **concentration of 17α-hydroxyprogesterone (17-OHP) in blood**.
   - Newborn blood is collected by heel stick onto blotter paper
   - 3 mm punches are taken from the blotted blood and analyzed for 17-OHP by enzyme-linked immunosorbent assay (ELISA)
   - Elevated 17-OHP concentration (definitions differ by municipality and birth weight) require follow-up testing
   - ***NOTE***: 17-OHP is elevated in both 21-hydroxylase deficiency and 11β-

hydroxylase deficiency, but **NOT** in 17α-hydroxylase deficiency.

2. ACTH stimulation test
    - An ACTH stimulation test is conducted by administering synthetic ACTH intravenously and monitoring peripheral patient blood for 17-OHP production by ELISA. Patients with normal 17α-hydroxylase activity will produce 17-OHP in response to ACTH, but patients with 17α-hydroxylase deficiency will maintain low serum 17-OHP concentration despite ACTH administration.

Case 2 (continued): ACTH stimulation test showed no increase in 17-OHP after intravenous administration of synthetic ACTH when compared with the basal level, suggesting an impairment in the pathway catalyzed by 17α-hydroxylase. Genetic analysis showed a compound heterozygous mutation in exon 6 of the *CYP17A1* gene (c.985-987TAC>AA and c.1118A>T). After being diagnosed with 17α-hydroxylase deficiency, the patient stopped taking antihypertensive medication and started oral hydrocortisone. The patient's blood pressure normalized without antihypertensive therapy, and follow up laboratory investigations revealed normal levels of ACTH (11.04 μg/dL), progesterone (0.04 ng/mL) and serum electrolytes.

(Clin Exp Reprod Med. 2015 Jun; 42(2): 72–76.

Answers to Study Questions

1. How is tissue-specific production of aldosterone, cortisol and DHEA achieved in the adrenal cortex?

Answer: Tissue-specific production of aldosterone is achieved by selective expression of AGTR1 and aldosterone synthase in the zona glomerulosa. Tissue-specific production of cortisol is achieved by selective expression of 17α-hydroxylase and 11β-hydroxylase in the zona fasciculata. Tissue specific production of DHEA is achieved by selective expression of cytochrome b5 in the zona reticularis.

2. Compare and contrast the hormonal regulation of aldosterone, cortisol, and DHEA production in the adrenal cortex.

Answer: Aldosterone production in the zona glomerulosa is regulated by angiotensin II, part of the RAA system that responds to drops in blood pressure. Cortisol production in the zona fasciculata is regulated by ACTH, part of the hypothalamus-pituitary-adrenal axis. Aldosterone does not directly feed back on the secretion of renin or angiotensin; rather, aldosterone promotes salt and water retention in the kidney, which leads to increased blood pressure. By contrast, cortisol directly feeds back on the hypothalamus and pituitary to suppress secretion of CRH and ACTH. DHEA production is stimulated by ACTH, but this steroid does NOT feedback on the secretion of CRH or ACTH. Both angiotensin II and ACTH bind GPCRs in their respective tissues, but the coupled heterotrimeric G protein is different and sets off different signal transduction cascades. However, for both stimuli, short-term stimulation leads to rapid increase in steroid production and long-term stimulation leads to increased cell proliferation and CAH.

3. By what biochemical mechanisms do angiotensin II and ACTH upregulate aldosterone and cortisol production in the short term (minutes)? Describe what happens with long-term stimulation of these hormones?

*Answer: **Short-term** receptor activation leads to phosphorylation of the steroid acute regulatory protein (StAR), promoting the translocation of cholesterol from the outer mitochondrial membrane to the inner mitochondrial membrane. Cholesterol side-chain cleavage takes place in the inner mitochondrial membrane. Consequently, the resulting increase in cholesterol bioavailability in the inner mitochondrial membrane enhances steroid hormone production 7- to 10-fold.*

***Long-term** receptor activation leads to activation of MAP kinase and other proliferative pathways. Chronic MAP kinase activation stimulates hypertrophy and hyperplasia of adrenal tissue.*

4. Describe the synthesis and regulation of testosterone, estrogen and progesterone in the testes and ovaries.

Steroidogenesis in the Leydig cells of the testes and the theca cells of the ovary is initiated by gonadotropin releasing hormone (GnRH) production in the hypothalamus, which stimulates production of luteinizing hormone (LH) and follicle stimulating hormone (FSH) in the anterior pituitary. LH binding to LH receptors (Gs-coupled GPCRs) in Leydig and theca cells leads to phosphorylation of StAR and initiation of steroidogenesis. The difference between Leydig cells and theca cells is the expression level of 17β-hydroxysteroid dehydrogenase (17βOHD), which is why Leydig cells produce mostly testosterone while theca cells primarily produce androstenedione. Testosterone feeds back on the hypothalamus and pituitary to inhibit GnRH and LH production.

During the follicular phase of the menstrual cycle, granulosa cells are non-steroidogenic. FSH binding stimulates expression of aromatase, which converts androstenedione into estrone and testosterone into estradiol. Granulosa cells express higher levels of 17βOHD than theca cells in order to convert estrone into estradiol. Estradiol feeds back to inhibit GnRH production by the hypothalamus and LH production by the pituitary. Granulosa cells produce inhibin to feedback regulate FSH production by the anterior pituitary.

In the luteal phase of the menstrual cycle, LH concentration spikes, which enables binding to low affinity LH receptors on granulosa cells. Synergistic LH and FSH receptor binding stimulates StAR activity and progesterone production, which helps to prepare the uterus for implantation of a fertilized ovum. Progesterone is a negative feedback regulator of the hypothalamus and pituitary.

5. Compare and contrast the pathogenesis related to mineralocorticoid, glucocorticoid and androgen effects in 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

*Answer: In **21-hydroxylase deficiency**, production of both aldosterone and cortisol is hindered or completely blocked. When completely blocked, lack of aldosterone will result in a salt-wasting syndrome due to the inability to stimulate sodium retention in the distal nephron. Consequently, newborns develop life-threatening hyponatremia, hyperkalemia and hypovolemic **hypotension**. The lack of cortisol will also lead to **hypoglycemia**. The resulting chronic increases in renin and ACTH due to low aldosterone and cortisol will lead to congenital adrenal hyperplasia (CAH). The presence of increased steroid hormone precursors progesterone and 17OHP plus the high levels of ACTH cause excess production of adrenal androgens from the fetal zone (and later, the zona reticularis), resulting in **virilization** of XX patients. Less severe forms of 21-hydroxylase deficiency are observed depending on the residual activity of the mutant enzyme.*

*In **11β-hydroxylase deficiency**, the production of cortisol **and** corticosterone are compromised, so patients are **hypoglycemic**, but the capacity to synthesize aldosterone remains intact. Decreased serum cortisol leads to chronically elevated ACTH levels, resulting in hyperplasia of the zona fasciculata. Synthetic intermediates such as deoxycorticosterone (DOC) and deoxycortisol accumulate, which have mineralocorticoid activity. The resulting stimulation of the mineralocorticoid receptor leads to unregulated renal sodium retention and **hypertension**. Note: although aldosterone production is unaffected by 11β-hydroxylase deficiency, the unregulated mineralocorticoid activity of DOC and deoxycortisol suppress renin release from the kidney, thus aldosterone levels are **LOW** in 11β-hydroxylase deficiency. The presence of increased steroid hormone precursors progesterone and 17OHP plus the high levels of ACTH cause excess production of adrenal androgens from the fetal zone (and later, the zona reticularis), resulting in **virilization** of XX patients.*

***17α-hydroxylase deficiency** results in a loss of cortisol, which can be partially made up by elevated corticosterone levels when ACTH levels are high. The presence of high levels of steroid precursor DOC from the*

high levels of ACTH stimulate salt retention in the kidney and consequent hypertension. The hypertension keeps renin levels low, leading to low levels of aldosterone (just like in 11β-hydroxylase deficiency). Androgen and estrogen production can be compromised, depending on the severity of the deficiency. In severe cases, XY patients can present with female-appearing secondary sexual characteristics. Neither XY nor XX patients will progress through puberty. Notably, some intermediates of the steroid biosynthetic pathway (notably deoxycortisol and 17OHP – see question 6) are underproduced in 17α-hydroxylase deficiency relative to normal or 11β-hydroxylase deficiency.

6. Summarize how 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency can be distinguished through laboratory testing.

Answer: 21-hydroxylase deficiency and 11β-hydroxylase deficiency both lead to a finding of elevated 17α-hydroxyprogesterone (17-OHP) in routine newborn blood screening panels in the United States and 50 other countries or municipalities. Although, in principle, these two most common presentations of congenital adrenal hyperplasia (CAH) might be distinguished through further analyte testing (especially by determining the concentration of deoxycorticosterone), 21-hydroxylase deficiency and 11β-hydroxylase deficiency are now typically distinguished via genotyping. In 17α-hydroxylase deficiency, 17-OHP levels are low, so newborns with these disorders are not identified as having CAH in routine newborn screening tests. Newborns may be identified by the presence of a difference in sexual development. Diagnosis is confirmed by failure to produce 17-OHP in an ACTH stimulation test and by genotyping.

Disorder	Complaint	DSD	Newborn Screen	ACTH challenge
21OHD (> 90%)	Hypotension	Virilizing (in XX)	High 17-OHP	Preg + 17-OHP +++ Cortisol – DHEA +++
11βOHD (< 10%)	Hypertension	Virilizing (in XX)	High 17-OHP	Preg + 17-OHP ++ Cortisol – DHEA +++
17αOHD (< 1%)	Hypertension	Non-virilizing* (XY with female characteristics)	Low (normal) 17-OHP	Preg + 17-OHP – Cortisol – DHEA –